

# evaluate\_external

April 7, 2026

## 1 External Validation: CLAM-MB + UNI2-h PAM50 Pipeline

This report evaluates the **CLAM-MB + UNI2-h** pipeline on two external cohorts, **without retraining or domain adaptation**:

- **HSI-BC** – 47 patients, one H&E .mrxs WSI per patient, IHC-derived subtype labels.
- **NOU** – 28 patients, 203 tissue-block H&E crops total, IHC-derived subtype labels.

The 10 CLAM-MB fold-checkpoints from TCGA-BRCA are reused as-is. Ensemble predictions average softmax probabilities across all 10 folds.

**PAM50 classes:** LumA (0), LumB (1), Basal (2), Her2 (3).

Both cohorts use **IHC surrogate** annotations, which introduces ~10–20% label noise vs. genomic PAM50.

### 1.1 Pipeline Overview

The pipeline is identical to the one in `evaluate_pam50.pdf`:

1. **Data preparation** – IHC surrogate labels mapped to PAM50 classes.
2. **Tissue patching** – CLAM `create_patches_fp.py`. HSI-BC: 256px patches from .mrxs; NOU: 512px patches from TIFF crops.
3. **Feature extraction** – UNI2-h (ViT-Giant/14, 1536-dim embeddings).
4. **Ensemble inference** – 10 TCGA-BRCA CLAM-MB checkpoints, mean-softmax ensemble. No fine-tuning or domain adaptation.

### 1.2 Experimental setup

[HSI-BC] loaded 47 ensemble predictions from 47 patients

[NOU] loaded 203 ensemble predictions from 28 patients

#### 1.2.1 Cohort Characteristics

Table 1: Cohort composition by PAM50 class

| =====                            |        |        |          |           |           |           |
|----------------------------------|--------|--------|----------|-----------|-----------|-----------|
| =====                            |        |        |          |           |           |           |
|                                  | Cohort | Slides | Patients | LumA      | LumB      | Basal     |
| Her2                             |        |        |          |           |           |           |
| TCGA-BRCA (training, 10-fold CV) |        | 986    | ~930     | 508 (52%) | 222 (23%) | 173 (18%) |
| 83 (8%)                          |        |        |          |           |           |           |

|          |                   |     |    |           |          |        |
|----------|-------------------|-----|----|-----------|----------|--------|
|          | HSI-BC (external) | 47  | 47 | 10 (21%)  | 29 (62%) | 4 (9%) |
| 4 (9%)   |                   |     |    |           |          |        |
|          | NOU (external)    | 203 | 28 | 152 (75%) | 23 (11%) | 6 (3%) |
| 22 (11%) |                   |     |    |           |          |        |
| =====    |                   |     |    |           |          |        |
| =====    |                   |     |    |           |          |        |

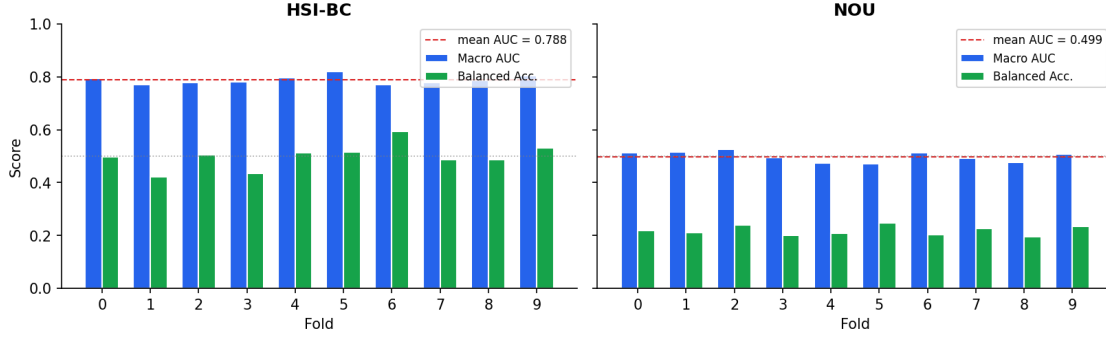
### 1.3 Per-Fold Ensemble Member Performance

Each of the 10 TCGA-trained checkpoints is applied independently to every external slide (no fold-wise splitting on the external set).

Table 2: Per-fold ensemble member metrics on each external cohort

| =====          |                       |                       |                        |  |  |  |
|----------------|-----------------------|-----------------------|------------------------|--|--|--|
| --- HSI-BC --- |                       |                       |                        |  |  |  |
|                | AUC                   | Acc                   | BAcc                   |  |  |  |
| Fold           |                       |                       |                        |  |  |  |
| 0              | 0.7938                | 0.3617                | 0.4978                 |  |  |  |
| 1              | 0.7708                | 0.2553                | 0.4220                 |  |  |  |
| 2              | 0.7788                | 0.3830                | 0.5065                 |  |  |  |
| 3              | 0.7809                | 0.2979                | 0.4345                 |  |  |  |
| 4              | 0.7976                | 0.3617                | 0.5142                 |  |  |  |
| 5              | 0.8196                | 0.4468                | 0.5159                 |  |  |  |
| 6              | 0.7705                | 0.5957                | 0.5927                 |  |  |  |
| 7              | 0.7778                | 0.4681                | 0.4871                 |  |  |  |
| 8              | 0.7865                | 0.2979                | 0.4884                 |  |  |  |
| 9              | 0.8050                | 0.4043                | 0.5315                 |  |  |  |
| Mean +/- Std:  | AUC=0.7881 +/- 0.0158 | Acc=0.3872 +/- 0.0992 | BAcc=0.4991 +/- 0.0480 |  |  |  |
| --- NOU ---    |                       |                       |                        |  |  |  |
|                | AUC                   | Acc                   | BAcc                   |  |  |  |
| Fold           |                       |                       |                        |  |  |  |
| 0              | 0.5121                | 0.4877                | 0.2192                 |  |  |  |
| 1              | 0.5160                | 0.4089                | 0.2118                 |  |  |  |
| 2              | 0.5269                | 0.5517                | 0.2405                 |  |  |  |
| 3              | 0.4953                | 0.4926                | 0.2019                 |  |  |  |
| 4              | 0.4742                | 0.4039                | 0.2097                 |  |  |  |
| 5              | 0.4716                | 0.5961                | 0.2466                 |  |  |  |
| 6              | 0.5141                | 0.4138                | 0.2037                 |  |  |  |
| 7              | 0.4917                | 0.5369                | 0.2264                 |  |  |  |
| 8              | 0.4772                | 0.5320                | 0.1966                 |  |  |  |
| 9              | 0.5088                | 0.5665                | 0.2358                 |  |  |  |
| Mean +/- Std:  | AUC=0.4988 +/- 0.0196 | Acc=0.4990 +/- 0.0698 | BAcc=0.2192 +/- 0.0174 |  |  |  |
| =====          |                       |                       |                        |  |  |  |

Figure 1: Per-fold ensemble member performance on external cohorts



#### 1.4 Aggregated Ensemble Predictions (Slide-Level)

Mean-softmax ensemble of 10 fold-checkpoints. For HSI-BC, slide-level = patient-level (one WSI per patient). NOU patient-level aggregation follows below.

Table 3: Aggregated ensemble metrics (slide-level)

| Cohort | Macro AUC | Acc    | BAcc   | Macro F1 | Wt. F1 |
|--------|-----------|--------|--------|----------|--------|
| HSI-BC | 0.8042    | 0.4043 | 0.5315 | 0.3905   | 0.4004 |
| NOU    | 0.5028    | 0.5222 | 0.2215 | 0.2250   | 0.5600 |

Note: BAcc (balanced accuracy) is the macro-averaged recall and equals random-chance level ( $1/4 = 0.25$ ) when the model collapses to a single class.

Classification report -- HSI-BC (slide-level)

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| LumA         | 0.33      | 0.60   | 0.43     | 10      |
| LumB         | 0.73      | 0.28   | 0.40     | 29      |
| Basal        | 0.50      | 0.25   | 0.33     | 4       |
| Her2         | 0.25      | 1.00   | 0.40     | 4       |
| accuracy     |           |        | 0.40     | 47      |
| macro avg    | 0.45      | 0.53   | 0.39     | 47      |
| weighted avg | 0.58      | 0.40   | 0.40     | 47      |

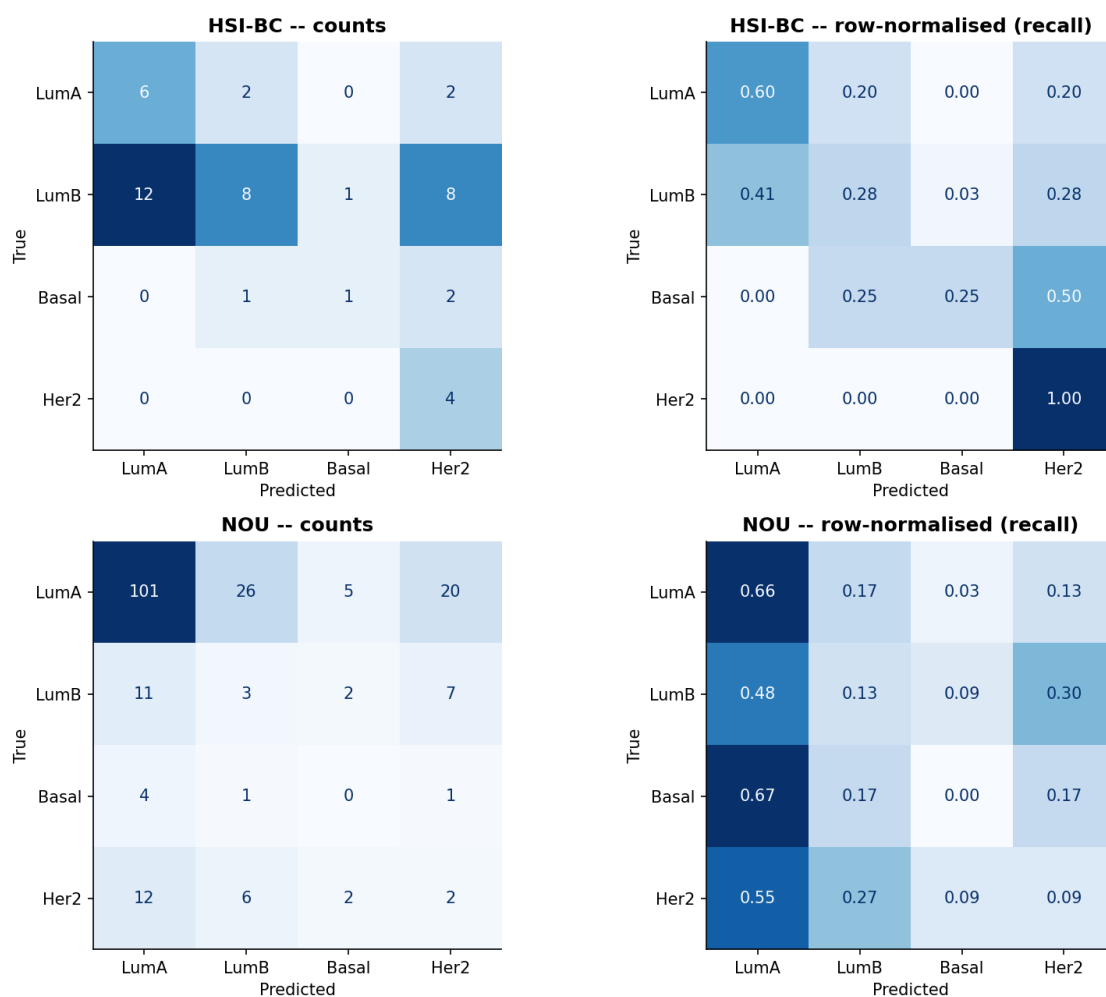
Classification report -- NOU (slide-level)

|  | precision | recall | f1-score | support |
|--|-----------|--------|----------|---------|
|--|-----------|--------|----------|---------|

|              |      |      |      |     |
|--------------|------|------|------|-----|
| LumA         | 0.79 | 0.66 | 0.72 | 152 |
| LumB         | 0.08 | 0.13 | 0.10 | 23  |
| Basal        | 0.00 | 0.00 | 0.00 | 6   |
| Her2         | 0.07 | 0.09 | 0.08 | 22  |
| accuracy     |      |      | 0.52 | 203 |
| macro avg    | 0.23 | 0.22 | 0.23 | 203 |
| weighted avg | 0.61 | 0.52 | 0.56 | 203 |

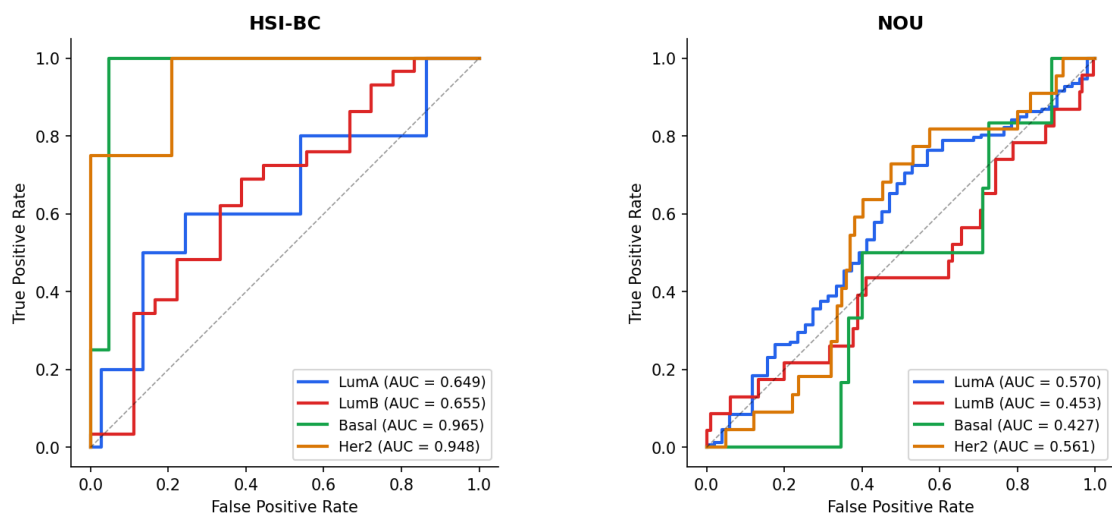
### 1.4.1 Confusion matrices

**Figure 2: Slide-level confusion matrices on external cohorts**



## 1.4.2 ROC Curves (One-vs-Rest)

Figure 3: One-vs-rest ROC curves on external cohorts



## 1.4.3 Per-class breakdown

Table 4: Per-class metrics on external cohorts (slide-level)

| Cohort | Class | Support | AUC    | Precision | Recall | F1     |
|--------|-------|---------|--------|-----------|--------|--------|
| HSI-BC | LumA  | 10      | 0.6486 | 0.3333    | 0.6000 | 0.4286 |
| HSI-BC | LumB  | 29      | 0.6552 | 0.7273    | 0.2759 | 0.4000 |
| HSI-BC | Basal | 4       | 0.9651 | 0.5000    | 0.2500 | 0.3333 |
| HSI-BC | Her2  | 4       | 0.9477 | 0.2500    | 1.0000 | 0.4000 |
| NOU    | LumA  | 152     | 0.5702 | 0.7891    | 0.6645 | 0.7214 |
| NOU    | LumB  | 23      | 0.4529 | 0.0833    | 0.1304 | 0.1017 |
| NOU    | Basal | 6       | 0.4272 | 0.0000    | 0.0000 | 0.0000 |
| NOU    | Her2  | 22      | 0.5608 | 0.0667    | 0.0909 | 0.0769 |

## 1.5 NOU Patient-Level Evaluation

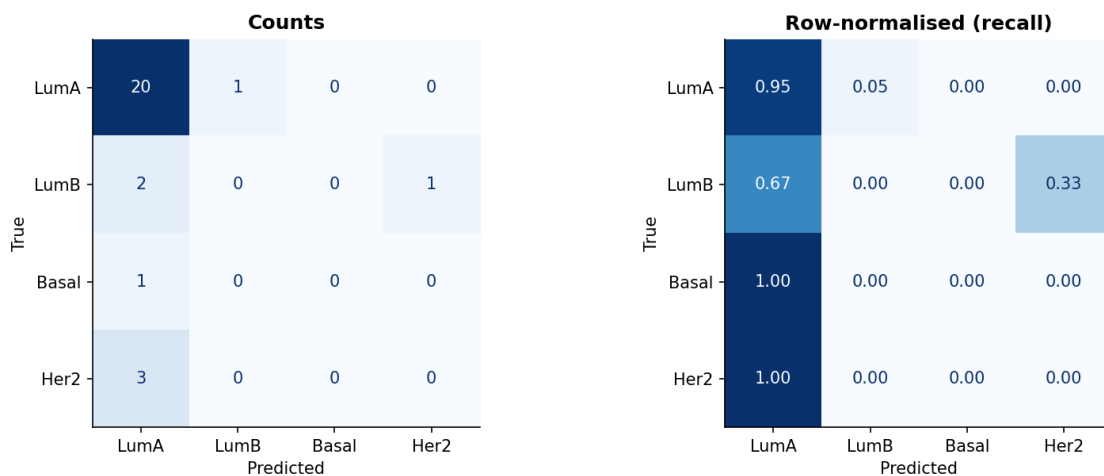
We aggregate the ensemble softmax probabilities across all tissue blocks of a patient by averaging, then argmax for the patient-level prediction.

NOU patient-level evaluation: n = 28 patients, median 7 blocks/patient

|                    |        |
|--------------------|--------|
| Macro AUC:         | 0.4246 |
| Accuracy:          | 0.7143 |
| Balanced accuracy: | 0.2381 |
| Macro F1:          | 0.2128 |
| Weighted F1:       | 0.6383 |

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| LumA         | 0.77      | 0.95   | 0.85     | 21      |
| LumB         | 0.00      | 0.00   | 0.00     | 3       |
| Basal        | 0.00      | 0.00   | 0.00     | 1       |
| Her2         | 0.00      | 0.00   | 0.00     | 3       |
| accuracy     |           |        | 0.71     | 28      |
| macro avg    | 0.19      | 0.24   | 0.21     | 28      |
| weighted avg | 0.58      | 0.71   | 0.64     | 28      |

**Figure 4: NOU patient-level confusion matrix (n=28)**



## 1.6 Comparison: Internal vs. External

Table 5: TCGA-BRCA internal vs. HSI-BC and NOU external

|                     | TCGA-BRCA (10-fold CV) | HSI-BC (slide=patient) | NOU (slide-level) |
|---------------------|------------------------|------------------------|-------------------|
| NOU (patient-level) |                        |                        |                   |
| Macro AUC           | 0.8879                 | 0.8042                 | 0.5028            |
| 0.4246              |                        |                        |                   |
| Accuracy            | 0.7181                 | 0.4043                 | 0.5222            |
| 0.7143              |                        |                        |                   |
| Balanced Acc.       | NaN                    | 0.5315                 | 0.2215            |
| 0.2381              |                        |                        |                   |
| Weighted F1         | 0.7179                 | 0.4004                 | 0.5600            |
| 0.6383              |                        |                        |                   |
| LumA AUC            | 0.8967                 | 0.6486                 | 0.5702            |

|           |        |        |        |
|-----------|--------|--------|--------|
| 0.6599    |        |        |        |
| LumB AUC  | 0.8174 | 0.6552 | 0.4529 |
| 0.4667    |        |        |        |
| Basal AUC | 0.9529 | 0.9651 | 0.4272 |
| 0.1852    |        |        |        |
| Her2 AUC  | 0.8846 | 0.9477 | 0.5608 |
| 0.3867    |        |        |        |
| =====     |        |        |        |
| =====     |        |        |        |

TCGA values are reproduced from evaluate\_pam50.pdf (10-fold CV, aggregated test set).

## 1.7 Findings

1. **HSI-BC: useful transfer, substantial drop.** Macro AUC drops from 0.888 (TCGA internal) to **0.804** (~9 pp). Per-class AUC ranking preserved (Basal, Her2, LumA, LumB). Low accuracy (0.40) reflects LumB dominance (62% of cohort) – the model frequently confuses LumB with LumA.
2. **NOU: model collapses to chance.** Slide-level macro AUC = 0.50, patient-level = 0.42. The 71% patient-level accuracy is misleading – it merely reflects LumA prevalence (75%). Balanced accuracy is ~0.24, i.e. random. All 10 ensemble members produce AUC in [0.47, 0.53], confirming systematic failure.
3. **NOU collapse is likely a domain shift issue.** NOU TIFFs are small tissue crops (~210 patches/slide vs. ~48k for HSI-BC). CLAM attention, trained on large bags, cannot extract signal from bags two orders of magnitude smaller.
4. **Class imbalance amplifies AUC–accuracy gap.** Both cohorts are heavily luminal-skewed (HSI-BC 83%, NOU 89%), with 4 or fewer Basal/Her2 patients each. Per-class metrics have wide uncertainty at these sample sizes.
5. **IHC-to-PAM50 label noise caps performance.** Both cohorts use IHC surrogates (~80% concordance with genomic PAM50), so some “errors” are label-definition disagreements.

## 1.8 Next Steps

1. **Re-patch NOU** with a patching strategy that matches the TCGA training distribution (larger bags, proper field-of-view alignment).
2. **Stain normalisation** (Macenko/Vahadane) before feature extraction to reduce domain shift.
3. **Bootstrap confidence intervals** (stratified, 1000 resamples) for all headline metrics.
4. **Light fine-tuning** of CLAM-MB on external data to distinguish feature-level vs. aggregator-level failure.
5. **Multimodal fusion** using clinico-pathological features (HSI-BC clinical data, NOU CTC EP/EMT counts) alongside WSI embeddings.

## 1.9 Summary

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CLAM-MB + UNI2-h | PAM50 Subtyping | External Validation

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Model: 10x checkpoints from TCGA-BRCA (no retraining)
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--- TCGA-BRCA (internal benchmark from evaluate_pam50.pdf) ---
Macro AUC      0.8879
Accuracy       0.7181
Weighted F1    0.7179
LumA AUC       0.8967
LumB AUC       0.8174
Basal AUC      0.9529
Her2 AUC       0.8846

--- HSI-BC (slide-level, n = 47) ---
Macro AUC      0.8042
Accuracy       0.4043
Balanced Acc.  0.5315
Weighted F1    0.4004

--- NOU (slide-level, n = 203) ---
Macro AUC      0.5028
Accuracy       0.5222
Balanced Acc.  0.2215
Weighted F1    0.5600

--- NOU (patient-level, n = 28) ---
Macro AUC      0.4246
Accuracy       0.7143
Balanced Acc.  0.2381
Weighted F1    0.6383
=====

```